

VIGNESH SUBRAMANIAN

Atlanta, Georgia, USA

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RESEARCH INTERESTS

I am a Ph.D. student at Georgia Tech working on building reinforcement learning systems that can both generalize reliably and provide clear, verifiable guarantees about their behavior. My research combines insights from formal methods, neuro-symbolic reasoning, and large language models to design agents that not only learn effectively but can also justify and certify their decisions in safety-critical settings.

EDUCATION

Georgia Institute of Technology

Ph.D. in Computer Science | 3rd Year | GPA: 3.9

Advisor – Prof. Suguman Bansal | *Anticipated Graduation: May 2028*

Aug 2023 – Ongoing

Atlanta, Georgia

National Institute of Technology, Tiruchirappalli

B.Tech. - Electrical and Electronics Engineering

Minor - Computer Science Engineering

Aug 2019 - April 2023

Tiruchirappalli, Tamil Nadu

Relevant Coursework: Logic in Computer Science, SAT/SMT solvers, Machine Learning and Deep Learning, Software Analysis and Testing, Artificial Intelligence, Data Structures and Algorithms

PUBLICATIONS

• CONFERENCES

Inductive Generalization in Reinforcement Learning from Specifications

Automated Technology for Verification and Analysis, 2025

Vignesh Subramanian, Rohit Kushwah, Subhajit Roy, Suguman Bansal

[\[Link\]](#)

Certification-Guided Evaluation of Reinforcement Learning Generalization (Presentation Only)

Symposium on AI Verification, 2025

Vignesh Subramanian, Djordje Zikelic, Suguman Bansal

[\[Link\]](#)

Reinforcement Learning for Stochastic Max-Plus Linear Systems

Conference on Decision and Control, IEEE 2023

Vignesh Subramanian, Farzaneh Farhadi, Sadegh Soudjani

[\[Link\]](#)

A Novel Facial Emotion Recognition Model Using Segmentation VGG-19 Architecture

International Journal of Information Technology, Springer, 2023

Vignesh Subramanian, Savithadevi, M. Sridevi, Rajeswari Sridhar

[\[Link\]](#)

SIHeDA-Net: Sensor to Image Heterogeneous Domain Adaptation for Sign Language Detection

Medical Imaging with Deep Learning, 2022

Ishikaa Lunawat, **Vignesh Subramanian**, S. P. Sharan

[\[Link\]](#)

• WORKSHOPS

Certification-Guided Evaluation of Reinforcement Learning Generalization (Short Paper)

AAAI - Workshop on Generalization in Planning, 2025

Vignesh Subramanian, Djordje Zikelic, Suguman Bansal

[\[Link\]](#)

Inductive Generalization in Reinforcement Learning from Specifications (Short Paper)

NeurIPS - Workshop on Generalization in Planning, 2023

Rohit Kushwah, **Vignesh Subramanian**, Subhajit Roy, Suguman Bansal

[\[Link\]](#)

TEACHING EXPERIENCE

CS 8803 - Formal Methods in Reinforcement Learning

Fall 2025

Georgia Institute of Technology

Atlanta, Georgia

- Assisting instruction and research-oriented discussions covering formal verification, neuro-symbolic reasoning, safe RL, correctness guarantees, and specification-guided learning.
- Supporting student final projects, literature reviews, and research direction selection at the intersection of reinforcement learning, formal methods, and trustworthy AI.

CS 8803 - SAT/SMT Solvers

Fall 2024

Georgia Institute of Technology

Atlanta, Georgia

- Guiding students on advanced topics in SAT and SMT solvers, including model checking, symbolic execution, neural network verification, and formal verification techniques.
- Supporting assignments and projects that integrate LLM-based reinforcement learning, LLM inference, and proof synthesis for solving complex problems in solver architecture and proof complexity.

INTERNSHIPS

Newcastle University, United Kingdom | Paper link

Jan 2022 - Nov 2022

Research Intern - Mentor: Prof. Sadegh Soudjani

Newcastle, England

- Proposed a novel optimization strategy using Deep Q-Learning for Stochastic Max-Plus-Linear Discrete Event Systems under uncertainties, achieving a 2.5x speedup over Model Predictive Control for minimizing stochastic delays in railway systems, published at **Conference on Decision and Control, IEEE, 2023**.

Samsung R&D Institute, Bangalore | Report link

Aug 2021 - Feb 2022

Research Intern - Mentor: Mr. Sujoy Saha, Mr. Rajat Kumar Jain

Karnataka, India

- Developed a Tri-Stage Occlusion Handling Normal Map Estimation Algorithm to enhance 3D human normal map estimation on blurred, noisy, low-res 2D images, achieving 92.78% IoU (vs. 71.26% baseline) by effectively addressing occlusions from shadows, blur, and background.

National Institute of Technology, Tiruchirappalli | Paper link | Github link

Feb 2021 - Dec 2021

Research Intern - Mentor: Prof. M. Sridevi

Tamil Nadu, India

- Designed a novel CNN architecture integrating U-Net and VGG layers for Facial Emotion Recognition, achieving 75.97% accuracy on the FER-2013 dataset, ranking in the **Top 5** on the global leaderboard. Findings were published in the **International Journal of Information Technology, Springer**.

TECHNICAL SKILLS

- **Languages:** C/C++, Python, Java, JavaScript, Lean
- **Packages/Frameworks:** PyTorch, Tensorflow, Keras, Scikit-learn, OpenCV, OpenAI Gym, Stable Baselines
- **Reinforcement Learning:** Policy Gradient Methods, Generalization Algorithms, Safety Certificates for RL, Reachability and Safety Analysis
- **Formal Verification:** Model Checking, Formal Methods for Safety Verification
- **LLM:** Pre-training and fine-tuning large-scale transformer models, LLM-based inference for complex logical queries, reinforcement learning from human feedback (RLHF), and automated proof generation.

POSITIONS OF RESPONSIBILITY

Reviewer

2025

AAAI — Reviewer for the AAAI Workshop on Generalization in Planning.

Student Research Mentor, Machine Learning and AI

2020 – 2023

Spider R&D, NIT Trichy — Led ML/AI projects; built SIHeDA-NET (Paper) and GISiL (Demo).

Coordinator, Workshops and Publicity

2020 – 2021

Currents Symposium, NIT Trichy — Managed workshops, publicity; hosted 600+ participants.